

Understanding .lib Files

INTRODUCTION:

This guide provides an introduction to reading and understanding a Liberty file (.lib), which is a standard format used in electronic design automation (EDA) to describe the characteristics of a cell library. This file contains critical information about the timing, power, and physical properties of the logic gates and other components used in a digital circuit design.

OBJECTIVES:

After completing this lab, you will be able to:

- Identify the key sections within a Liberty file.
- Understand the purpose of different attributes and their values.
- Interpret timing and power data for a given cell.
- Locate information about pin characteristics and logical functions.

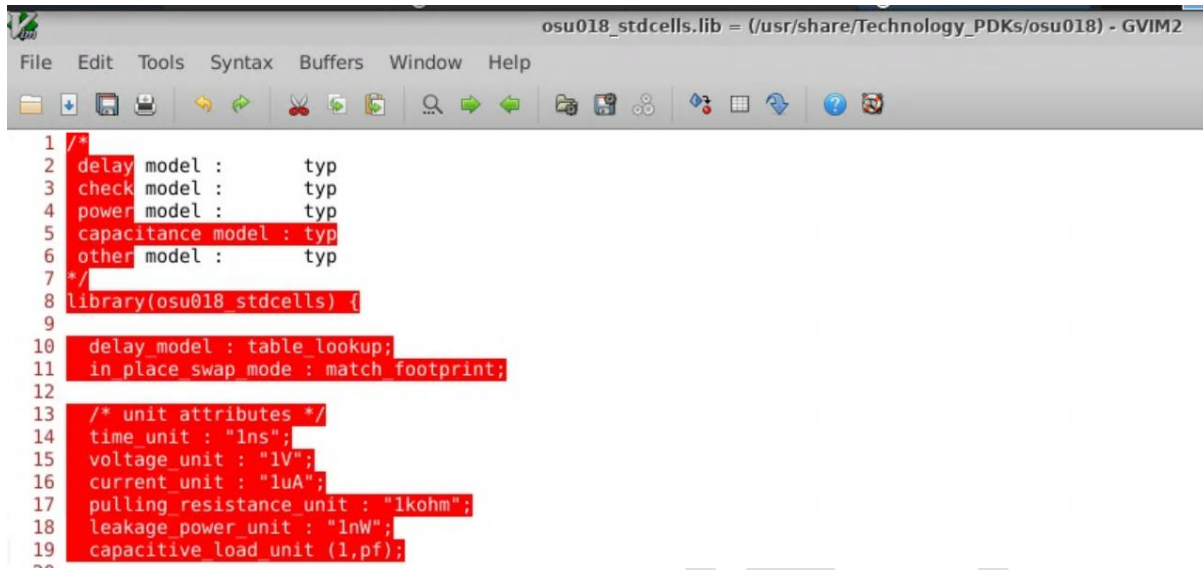
PROCEDURE:

A Liberty file is typically organized in a hierarchical structure, with different sections and attributes defining the library's properties and the individual cells within it.

This lab consists of the following steps:

Section 1: Library-Level Attributes The file begins with attributes that apply to the entire library. These may include:

- `library()`: The main block that encapsulates all library information.
- `technology`: Specifies the semiconductor technology, e.g., "cmos."
- `date`: The date the library was generated.
- `revision`: The revision number of the library.
- `delay_model`: The timing model used, e.g., "nldm" (Non-Linear Delay Model).
- `time_unit`, `voltage_unit`, `current_unit`, `capacitive_unit`: Defines the units for various measurements within the file.



```

1 /*
2 delay model : typ
3 check model : typ
4 power model : typ
5 capacitance model : typ
6 other model : typ
7 */
8 library(osu018_stdcells) {
9
10 delay_model : table lookup;
11 in_place_swap mode : match footprint;
12
13 /* unit attributes */
14 time_unit : "1ns";
15 voltage_unit : "1V";
16 current_unit : "1uA";
17 pulling_resistance_unit : "1kohm";
18 leakage_power_unit : "1nW";
19 capacitive_load unit (1,pf);
20

```

Section 2: Operating Conditions and Wire Load Models This section describes the conditions under which the library was characterized and the models used for interconnect parasitics.

- `operating_conditions()`: Defines parameters like process (process), voltage (voltage), and temperature (temperature).
- `wire_load_model()`: Provides a model to estimate the resistance and capacitance of interconnects based on fanout

Example:

```

operating_conditions("typical_conditions") {
    process : 1.0;
    voltage : 1.2;
    temperature : 25;
}

wire_load_model ("WLM_10X") {
    resistance : 0.1;
    capacitance : 0.05;
    fanout_length (1, 10);
    fanout_length (2, 20);
}

```

```
osu018_stdcells.lib = (/usr/share/Technology_PDKs/osu018) - GVIM2
File Edit Tools Syntax Buffers Window Help
29 nom_process : 1;
30 nom_voltage : 1.8;
31 nom_temperature : 25;
32 operating_conditions ( typical ) {
33     process : 1;
34     voltage : 1.8;
35     temperature : 25;
36 }
37 default_operating_conditions : typical;
```

Section 3: Cell Definitions The cell() block is the most detailed section, containing all the information for a specific logic gate or component.

- cell(cell_name): The main block for a single cell, e.g., cell(AND2X1).
- area: The physical area of the cell.
- ff or latch or nand, etc.: The functional definition of the cell.

```
osu018_stdcells.lib = (/usr/share/Technology_PDKs/osu018) - GVIM2
File Edit Tools Syntax Buffers Window Help
130 /* ----- *
131 * Design : AND2X1 *
132 * ----- */
133 cell (AND2X1) {
134     area : 32;
135     cell_leakage_power : 0.0746794;
136     pin(A) {
137         direction : input;
138         capacitance : 0.0129077;
139         rise_capacitance : 0.0129077;
140         fall_capacitance : 0.0128842;
141     }
142     pin(B) {
143         direction : input;
144         capacitance : 0.0125298;
145         rise_capacitance : 0.0125298;
146         fall_capacitance : 0.0122586;
147     }
148     pin(Y) {
149         direction : output;
150         capacitance : 0;
151         rise_capacitance : 0;
152         fall_capacitance : 0;
153         max_capacitance : 0.505476;
154         function : "(A & B)";
155         timing() {
156             related_pin : "A";
157             timing_sense : positive unate;
```

EXPLANATION:

- cell(AND2X1): Defines a cell named "AND2X1", which is a 2-input AND gate.
- area: The physical area of the cell is 10 (in whatever units were defined).
- pin(A) and pin(B): These are the two input pins. Each has an input capacitance of 1.
- pin(Z): This is the output pin. Its function is defined as "A & B".

CONCLUSION:

In this guide, you learned how to read and understand a Liberty file. By breaking down the file into its main sections and exploring the purpose of lookup tables, you can interpret the timing and power data for digital components. This knowledge is essential for working with EDA tools in circuit design.

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